Quiz 2

1)

Balanced flows :

inflows into unemployment: $s \cdot L$ = outflows from unemployment: $f(\theta) \cdot U$

2) $m = \omega \cdot U^{\eta} \cdot V^{1-\eta}$, $\theta = \frac{V}{U}$

From q.1 :

$$sL = f(\theta) \cdot U$$
$$sL = f(\theta) \cdot (H - L)$$
$$L(s + f(\theta)) = f(\theta) \cdot H$$
$$L = \frac{f(\theta) \cdot H}{s + f(\theta)}$$

, $f = \frac{m}{U}$

$$f = \frac{\omega \cdot U^{\eta} \cdot V^{1-\eta}}{U}$$
$$f = \omega \cdot U^{\eta - 1} \cdot V^{1-\eta}$$
$$f = \omega \cdot \theta^{1-\eta}$$

3) $L = \frac{\omega \cdot \theta^{1-\eta} \cdot H}{s + \omega \theta^{1-\eta} H}$

$$\frac{dL}{d\theta} = \frac{\omega(1-\eta) \cdot \theta^{-\eta} \cdot H \cdot (s + \omega \theta^{1-\eta} \cdot H) - (1-\eta) \omega \theta^{-\eta} \cdot H \cdot \omega \cdot \theta^{1-\eta} \cdot H}{(s + \omega \theta^{1-\eta} \cdot H)^2}$$

$$\frac{dL}{d\theta} = \frac{(1-\eta) \cdot \omega \cdot \theta^{-\eta} \cdot H \left[s + \cancel{\omega \theta^{1-\eta} \cdot H} - \cancel{\omega \cdot \theta^{1-\eta} \cdot H} \right]}{(s + \omega \theta^{1-\eta} \cdot H)^2} > 0 \quad (L \text{ is increasing in } \theta)$$

$$\frac{d^2 L}{d\theta^2} = \frac{-(1-\eta) \cdot \eta \cdot \omega \theta^{-\eta-1} \cdot H \cdot s \cdot (s + \omega \theta^{1-\eta} \cdot H)^2 - (1-\eta) \omega \theta^{-\eta} \cdot H \cdot s \cdot 2 (s + \omega \theta^{1-\eta} \cdot H) \cdot (1-\eta) \omega H \cdot \theta^{-\eta}}{(s + \omega \theta^{1-\eta} \cdot H)^4}$$

$$\frac{d^2 L}{d\theta^2} = \frac{-(1-\eta) \cdot \omega \cdot H \cdot s \cdot \theta^{-\eta} \left[\eta \cdot \theta^{-1} (s + \omega \theta^{1-\eta} \cdot H)^2 + 2 (s + \omega \theta^{1-\eta} \cdot H)(1-\eta) \omega H \theta^{-\eta} \right]}{(s + \omega \theta^{1-\eta} \cdot H)^4} < 0$$

$\rightarrow$ L is concave in θ

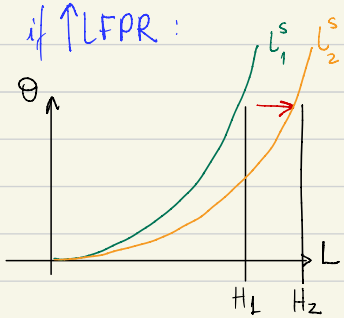
if $\theta = 0$, $f(\theta = 0) = 0$, $L = \dfrac{H}{\underbrace{\frac{s}{f(\theta)}}_{\infty} + 1} \to 0$

if $\theta = \infty$, $f(\theta = \infty) = \infty$, $L = \dfrac{H}{\underbrace{\frac{s}{f(\theta)}}_{0} + 1} = H$

4) $\uparrow\theta$ makes it easier to find jobs, and it's harder to fill a vacancy.

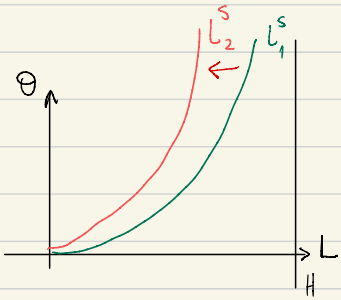
$\uparrow\theta = \dfrac{V\uparrow}{U\downarrow}$

5) if $\uparrow$LFPR :



6) $\uparrow s$

Recall that $L^s(\theta) = \dfrac{f(\theta)}{\uparrow s + f(\theta)} \cdot H \quad \to \downarrow L^s$



7) $m = \sqrt{U}\sqrt{V}$, $\theta = \frac{V}{U}$, $\tau = \frac{R}{N}$, $q = \frac{m}{V}$

$$q = \frac{\sqrt{U}\sqrt{V}}{V}$$

$$q = \sqrt{\frac{U}{V}}$$

$$q = \frac{1}{\sqrt{\theta}}$$

$$\tau(\theta) = \frac{r \cdot s}{q(\theta) - r \cdot s} = \frac{r \cdot s}{\frac{1}{\sqrt{\theta}} - r s} = \frac{r \cdot s}{\frac{1 - r \cdot s \sqrt{\theta}}{\sqrt{\theta}}} = \frac{r \cdot s \cdot \sqrt{\theta}}{1 - r \cdot s \sqrt{\theta}}$$

8) $\tau = \frac{r \cdot s \sqrt{\theta}}{1 - r s \sqrt{\theta}}$

$$\frac{d\tau}{d\theta} = \frac{\frac{1}{2} \cdot r \cdot s \cdot \theta^{-\frac{1}{2}} \cdot (1 - r \cdot s \sqrt{\theta}) - \left(-\frac{1}{2} \cdot r s \theta^{-\frac{1}{2}}\right) \cdot r \cdot s \sqrt{\theta}}{(1 - r \cdot s \sqrt{\theta})^2}$$

$$\frac{d\tau}{d\theta} = \frac{\frac{1}{2} \cdot r s \theta^{-\frac{1}{2}} \left[1 - \cancel{r \cdot s \sqrt{\theta}} + \cancel{r s \sqrt{\theta}}\right]}{(1 - r s \sqrt{\theta})^2} > 0 \quad (\tau \text{ is increasing in } \theta)$$

In order for τ to be positive, $1 - r \cdot s \sqrt{\theta} > 0$

$$\sqrt{\theta} < \frac{1}{r \cdot s}$$

$$\theta < \frac{1}{(r s)^2} \quad \therefore \quad \theta \in \left[0, \frac{1}{(r \cdot s)^2}\right]$$

9) $m = \sqrt{U}\cdot\sqrt{V}$, $y = 2\cdot a\sqrt{N}$, $\theta = \frac{V}{U}$

$L^d = ?$

Labor cost: $wL = w(R+N)$, recall that $R = T(\theta)\cdot N$

$$\pi(N) = 2\cdot a\sqrt{N} - wN(T(\theta)+1)$$

$$\pi'(N) = 0$$

$$\pi'(N) = \cancel{2}\cdot a\cdot\frac{1}{\cancel{2}}\cdot N^{-\frac{1}{2}} - w(T(\theta)+1) = 0$$

$$N^{-\frac{1}{2}} = \frac{w(T(\theta)+1)}{a}$$

$$N = \left(\frac{w^2(T(\theta)+1)^2}{a^2}\right)^{-1}$$

, recall that $L = N + \overset{T(\theta)\cdot N}{R} = N(1+T(\theta))$

$$(1+T(\theta))N = \left[\frac{a}{w(T(\theta)+1)}\right]^2\cdot(1+T(\theta))$$

$$L = \left(\frac{a}{w}\right)^2\cdot\frac{1}{T(\theta)+1}$$

, recall that $T(\theta)+1 =$

$$= \frac{r\cdot s\sqrt{\theta}}{1-r\cdot s\sqrt{\theta}} + \frac{1-r\cdot s\sqrt{\theta}}{1-r\cdot s\sqrt{\theta}}$$

$$T(\theta)+1 = \frac{1}{1-r\cdot s\sqrt{\theta}}$$

$$L^d = \left(\frac{a}{w}\right)^2\cdot(1-r\cdot s\sqrt{\theta})$$

10) $\frac{dL^d}{d\theta} = -\left(\frac{a}{w}\right)^2\cdot r\cdot s\cdot\frac{1}{2}\cdot\theta^{-\frac{1}{2}} < 0$ L decreases in θ.

$$L^d(\theta=0) = \left(\frac{a}{w}\right)^2$$

$$L^d\left(\theta = \frac{1}{(rs)^2}\right) = \left(\frac{a}{w}\right)^2\left(1 - \cancel{rs}\cdot\frac{1}{\cancel{rs}}\right) = 0$$

$\to 0$

11) $\uparrow a$

$$L^d = \left(\frac{a}{w}\right)^2 (1 - rs\sqrt{\theta})$$

$\uparrow a \rightarrow \uparrow L^d$

12)

$$L^d = \left(\frac{a}{w}\right)^2 (1 - rs\sqrt{\theta})$$

$\uparrow w \rightarrow \downarrow L^d$

$\uparrow \theta \rightarrow \downarrow L^d$